# Binomial Theorem: Formula Sheet

## 1. Basics

Theorem:

$$(x+y)^n = \sum_{r=0}^{n} {}^nC_r x^{n-r} y^r$$

Number of terms: $n+1$

Sum of binomial coefficients: $\sum {}^nC_r = 2^n$

Sum of coefficients: variable $= 1$

Symmetry: ${}^nC_r = {}^nC_{n-r}$

Pascal: ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$

Equal:

$${}^nC_x = {}^nC_y \Rightarrow x = y \text{ or } x + y = n$$

Ratio:

$$\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}$$

## 2. Terms

General term: $T_{r+1} = {}^nC_r x^{n-r} y^r$

$k$th term from the end: $T_{n-k+2}$

Consecutive terms:

$$\frac{T_{r+1}}{T_r} = \frac{n-r+1}{r} \cdot \frac{y}{x}$$

Middle term ($n$ even): $T_{\frac{n}{2}+1}$

Middle terms ($n$ odd):

$$T_{\frac{n+1}{2}},\ T_{\frac{n+3}{2}}$$

Greatest binomial coefficient: ${}^nC_{n/2}$ or ${}^nC_{(n\pm1)/2}$

NGT: $p = \dfrac{n+1}{1+|a/b|}$; $T_{[p]+1}$ ($p$ integer: $T_p, T_{p+1}$)

$T_k$ is NGT: $k-1 < p < k$

## 3. Multinomial

General term:

$$\frac{n!}{r_1! \cdots r_k!} x_1^{r_1} \cdots x_k^{r_k}$$

Condition: $r_1 + \cdots + r_k = n$

Dissimilar terms: ${}^{n+k-1}C_{k-1}$

## 4. Remainders and parts

$a^n - b^n$ divisible by $a-b$ for all $n$

$a^n - b^n$ divisible by $a+b$ for even $n$

$a^n + b^n$ divisible by $a+b$ for odd $n$

Mod $d^2$: $(1+kd)^n \equiv 1 + nkd$

$R = I + f,\ G = (a - \sqrt{b})^n$: $0 < G < 1 \Rightarrow I = R + G - 1$

$$RG = (a^2 - b)^n$$

## 5. Sums of series

Even coefficients:

$$\frac{f(1) + f(-1)}{2}$$

Odd coefficients:

$$\frac{f(1) - f(-1)}{2}$$

Alternating partial sum:

$$\sum_{r=0}^{k} (-1)^r {}^nC_r = (-1)^k {}^{n-1}C_k$$

Hockey stick:

$$\sum_{k=r}^{m} {}^kC_r = {}^{m+1}C_{r+1}$$

Remove $r$: $r\,{}^nC_r = n\,{}^{n-1}C_{r-1}$

$$\sum r\,{}^nC_r = n2^{n-1}$$

$$\sum r^2\,{}^nC_r = n(n+1)2^{n-2}$$

Integration:

$$\sum \frac{{}^nC_r}{r+1} = \frac{2^{n+1}-1}{n+1}$$

$$\frac{{}^nC_r}{r+1} = \frac{{}^{n+1}C_{r+1}}{n+1}$$

Sum pattern:

$$\sum {}^nC_r\,{}^mC_{k-r} = {}^{n+m}C_k$$

Difference pattern:

$$\sum {}^nC_r\,{}^mC_{r+k} = {}^{n+m}C_{n+k}$$

$$\sum ({}^nC_r)^2 = {}^{2n}C_n$$

GP:

$$(1+x)^m + \cdots + x^m = (1+x)^{m+1} - x^{m+1}$$

## 6. Any index and series

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \ldots$$

$$(1-x)^{-n} = 1 + nx + \frac{n(n+1)}{2!}x^2 + \ldots$$

Coefficient of $x^r$ in $(1-x)^{-n}$: ${}^{n+r-1}C_r$

Coefficient of $x^r$ in $(1+x)^{-n}$: $(-1)^r {}^{n+r-1}C_r$

Validity: $|x| < 1$

$$e^x = \sum \frac{x^n}{n!}$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \ldots$$

$(1+x)^n \approx 1 + nx$ for small $x$